Liquid stability, LLE and VLLE

After midterm · Session 3 of 6 · 180 minutes

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The question for today

When does a calculated liquid split represent stable equilibrium?

Before class: 3-page reading · 15–20 minutes

Bring your preparation answers. Start with the Week 2 clarification slides and Lab 08 guide for a 15-minute review. Then return here for stability and LLE.

The 180-minute class

ACTIVITY	MINUTES
Week 2 guided review	15
Stability question and essential concepts	15
One hand calculation	20
One lab demonstration	20
Break	10
Student pair investigation	70
Discussion, exit ticket and feedback	30

The 70-minute investigation is student work. Final block: discussion 20 min, exit ticket 5 min, feedback 3 min, closing 2 min.

Phase equilibrium includes stability

Equality of chemical potentials identifies stationary candidate states. Stable equilibrium requires the lowest admissible total Gibbs energy.

$$g_{mix}/RT = \sum_i x_i \ln x_i + g^E/(RT)$$

A split can lower G even when a homogeneous liquid is locally stable.

The common-tangent conditions

$$g'(x^\alpha) = g'(x^\beta) = \frac{g(x^\beta) - g(x^\alpha)}{x^\beta - x^\alpha}$$

The tangent must not lie above g(x) anywhere in the admitted composition interval.

A trivial root $x^\alpha = x^\beta$ does not establish two distinct liquid phases.

A lever-rule calculation

Synthetic tie line: $x_1^{\alpha}=0.2$, $x_1^{\beta}=0.8$. Feed $z_1=0.5$.

$$f_\beta = \frac{z_1 - x_1^\alpha}{x_1^\beta - x_1^\alpha} = 0.5$$

For $z_1=0.35$, $f^{\beta}=0.25$. At the same coexistence conditions, the phase amounts change but the tie-line endpoints do not.

An independent check

Use the calculated endpoints to close the component balance independently. Check that the tangent supports the entire admitted Gibbs curve.

One demonstration · 20 minutes

Open the core lab

In Lab 04 select a supported liquid split and save it. Change only overall composition inside the tie line. Compare phase compositions with phase fractions.

Keep one saved baseline for the pair investigation.

Pair investigation · 70 minutes

- 10 min: predict and record the baseline.
- 25 min: change one input and calculate.
- 20 min: exchange roles and check the result by hand.
- 15 min: prepare one plot or table and a short explanation.

In Lab 04, retain one supported tie line and compare two overall compositions inside it. Verify the lever rule by hand and explain the global stability check.

Discussion · 20 minutes

When does a calculated liquid split represent stable equilibrium?

Compare results and discuss unexpected behavior. Use one numerical check to support your explanation.

After class · 30–40 minutes

In Lab 04, retain one supported tie line and compare two overall compositions inside it. Verify the lever rule by hand and explain the global stability check.

Assessment 3 and independent checkpoint

Retain one study JSON, one comparison and a short explanation.

Optional: Session appendix.

Learning path · Offline package

Exit ticket + feedback · 8 minutes

5 min · Explain and check

Answer the session question in the form. Give one result or independent check from today’s lab, with units.

3 min · Help improve the next class

Rate the pace, report lab access, and suggest one improvement.

Thai or English. No name, ID or email requested.

Wait for **“Response received.”** We will use the final 2 minutes to close the class.

Scan to submit Session 3 exit ticket and feedback.

Open Session 3 form